3 Perverse Sheaf

3.1 Perverse t-stru.

<u>Def</u>: X var. perv. t-stru. on X, t-stru. on $D^b_c(X,\Bbbk)$ given by

$${}^pD^b_c(X,\Bbbk)^{\le 0} = \{\mathcal{F} \in D^b_c(X,\Bbbk) \mid \forall i \ \dim \operatorname{supp} H^i(\mathcal{F}) \le -i\}$$
$${}^pD^b_c(X,\Bbbk)^{\ge 0} = \{\mathcal{F} \in D^b_c(X,\Bbbk) \mid \forall i \ \operatorname{md supp} H^i(\mathbb{D}\mathcal{F}) \le -i\}$$

Thm: X var.

(1) $({}^pD^b_c(X,\Bbbk)^{\le 0}, {}^pD^b_c(X,\Bbbk)^{\ge 0})$ bounded t-struct. on $D^b_c(X,\Bbbk)$

(2) $(X_s)_{s\in\mathscr{S}}$ good stratification of X. $\Rightarrow$
$({}^pD^b_{\mathscr{S}}(X,\Bbbk)^{\le 0}, {}^pD^b_{\mathscr{S}}(X,\Bbbk)^{\ge 0})$ bounded t-struct. on $D^b_{\mathscr{S}}(X,\Bbbk)$

$\operatorname{Perv}(X,\Bbbk) = {}^pD^b_c(X,\Bbbk)^{\le 0} \cap {}^pD^b_c(X,\Bbbk)^{\ge 0}$

perverse sheaves.

(3.1.2): $\Bbbk$ field.

$\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\ge 0} \Leftrightarrow \mathbb{D}\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\le 0}$

$\Rightarrow \operatorname{Perv}(X,\Bbbk)$ is $\mathbb{D}$-inv. (Lem 3.1.11)

<u>Lem 3.1.3</u>: X smooth conn var $\Rightarrow$

$${}^pD^b_{locf}(X,\Bbbk)^{\le 0} = D^b_{locf}(X,\Bbbk)^{\le -\dim X}$$
$${}^pD^b_{locf}(X,\Bbbk)^{\ge 0} = D^b_{locf}(X,\Bbbk)^{\ge -\dim X}$$

$\mathscr{S}$ triv str.

$\operatorname{Perv}_{\mathscr{S}}(X,\Bbbk) = \operatorname{Loc}^{ft}(X,\Bbbk)[\dim X]$

<u>Lem 3.1.4</u>: $j: U \hookrightarrow X$ open embedding
$i: Z \hookrightarrow X$ closed embedding

(1) $j^* = j^!$ & $i_* = i_!$ t-exact

(2) $j_!$ & i^* sends ${}^pD_c^{b,\le 0}$ to ${}^pD_c^{b,\le 0}$

(3) j_* & $i^!$ sends ${}^pD_c^{b,\ge 0}$ to ${}^pD_c^{b,\ge 0}$

Pf: (1) follows from j^* & i_* · t-exact (for natural t-structure)
· comm w/. $\mathbb{D}$
· do not increase the sd / md. dim. of supp

(2) $j_!$ & i^* t-exact & do not increase the dim of supp.

(3) $\operatorname{md supp} H^i(\mathbb{D}(i^!\mathcal{F})) = \operatorname{md supp} H^i(i^*\mathbb{D}\mathcal{F}) \le \operatorname{md supp} H^i(\mathbb{D}\mathcal{F}) \le -i$
$\Rightarrow i^!\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\ge 0}$ □

Ex 3.1.1 $x \in X$. $i_x: \{x\} \hookrightarrow X$ inclusion map

$\operatorname{supp}^i\mathcal{F} = \overline{\{x \in X \mid H^i(i_x^*\mathcal{F}) \ne 0\}} \subset X$

${}^pD^b_c(X,\Bbbk)^{\ge 0} = \{\mathcal{F} \in D^b_c(X,\Bbbk) \mid \forall i \ \dim \operatorname{supp}^i\mathcal{F} \le -i\}$

$\operatorname{cosupp}^i\mathcal{F} = \overline{\{x \in X \mid H^i(i_x^!\mathcal{F}) \ne 0\}} \subset X$

$\operatorname{md supp}(\mathbb{D}\mathcal{F})$ $\quad x_s \in X_s$ smooth

$(\mathbb{D}_X\mathcal{F})_{x_s} = i_{x_s}^*\mathbb{D}_X\mathcal{F} = \mathbb{D}_{x_s} i_{x_s}^!\mathcal{F}$

$\operatorname{md supp}(\mathbb{D}\mathcal{F}) = \sup_{X_s}\{\dim X_s - \operatorname{grade}(\mathcal{F}|_{X_s})\}$

$\dim X_s - \operatorname{grade}(H^i(\mathbb{D}\mathcal{F}|_{X_s}))$
$= \dim(X_s) - \operatorname{grade}(H^i(i_{X_s}^*\mathbb{D}_X\mathcal{F}))$
$= \dim(X_s) - \operatorname{grade}(H^i(i_{x_s}^*\mathbb{D}_X\mathcal{F}))$
$= \dim(X_s) - \operatorname{grade}(H^i(\mathbb{D}_{X_s} i_{x_s}^!\mathcal{F}))$

• $D^b(\Bbbk\text{-mod}^{fg})^{\ge 0} = \{X \in D^b(\Bbbk\text{-mod}^{fg}) \mid \forall i \ \operatorname{grade} H^i(\mathbb{D}(X)) \ge i\}$

$\mathcal{F} \in {}^pD^{b,\ge 0}$
$\Leftrightarrow \operatorname{md supp} H^i(\mathbb{D}\mathcal{F}) \le -i$
$\Leftrightarrow \inf\{-\dim(X_s) + \operatorname{grade}(H^i(\mathbb{D}_{X_s} i_{x_s}^!\mathcal{F}))\} \ge i$
$\Leftrightarrow \inf\{-\dim(X_s) + \operatorname{grade}(H^i(\mathbb{D}_{X_s} i_{x_s}^!\mathcal{F}))\} \ge i$
$\Leftrightarrow \forall X_s \ni x_s \ \operatorname{grade}(H^i(\mathbb{D}_{X_s} i_{x_s}^!\mathcal{F})) \ge i + \dim X_s$
$\Leftrightarrow \forall X_s \ni x_s \quad i_{x_s}^!\mathcal{F} \in D^b(\Bbbk\text{-mod})^{\ge -\dim X_s}$
$\Leftrightarrow \dim \operatorname{cosupp}^i\mathcal{F} \le i$.

<u>Lem 3.1.5</u>: $j: U \hookrightarrow X$ open emb, $i: Z \hookrightarrow X$ closed emb. $\mathcal{F} \in D^b_c(X,\Bbbk)$

(1) $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\le 0} \Leftrightarrow j^*\mathcal{F} \in {}^pD^b_c(U,\Bbbk)^{\le 0}$, $i^*\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\le 0}$

(2) $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\ge 0} \Leftrightarrow j^!\mathcal{F} \in {}^pD^b_c(U,\Bbbk)^{\ge 0}$, $i^!\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\ge 0}$

"only if" immediate by 3.1.4

"if": $j_!j^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F} \to$.
$i_!i^!\mathcal{F} \to \mathcal{F} \to j_*j^*\mathcal{F} \to$.

<u>Lem 3.1.6</u>: X var. $(X_s)_{s\in\mathscr{S}}$ strat. $\forall s \in \mathscr{S}$ $j_s: X_s \hookrightarrow X$ inclu.
$\mathcal{F} \in D^b_c(X,\Bbbk)$.

(1) $\mathcal{F}$ constr. w.r.t. $\mathscr{S}$. We have $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\le 0}$
iff $j_s^*\mathcal{F} \in D^b_{locf}(X_s,\Bbbk)^{\le -\dim X_s}$ $\forall s \in \mathscr{S}$

(2) $\mathbb{D}\mathcal{F}$ constr. w.r.t. $\mathscr{S}$. We have $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\ge 0}$
iff $j_s^!\mathcal{F} \in D^b_{locf}(X_s,\Bbbk)^{\ge -\dim X_s}$ $\forall s \in \mathscr{S}$

• induction on # strata.

<u>Prop. 3.1.10</u>: $i_*: Z \hookrightarrow X$ inclusion of closed subvar.
i_* induces an equiv. of cat.
$\operatorname{Perv}(Z,\Bbbk) \xrightarrow{\sim} \{\mathcal{F} \in \operatorname{Perv}(X,\Bbbk) \mid \operatorname{supp}\mathcal{F} \subset Z\}$
& Right-hand side Serre subcat. of $\operatorname{Perv}(X,\Bbbk)$

<u>Prop 3.1.12</u>: $f: X \to Y$ finite morph.
$f_*: D^b_c(X,\Bbbk) \to D^b_c(Y,\Bbbk)$
t-exact. for the perverse t-structure

Pf: For lemma 2.3.19. f_* exact and preserves dim supp & md supp
f proper $\Rightarrow f_*$ comm w/. $\mathbb{D}$.

$\Rightarrow f_*{}^pD_c^{b,\ge 0} \subseteq {}^pD_c^{b,\ge 0}$
$f_*{}^pD_c^{b,\le 0} \subseteq {}^pD_c^{b,\le 0}$

3.2 Hom & $-\overset{L}{\otimes}-$ operator.

Lem 3.2.1. $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\le 0}$. $\mathcal{G} \in {}^pD^b_c(X,\Bbbk)^{\ge 0}$
$\Rightarrow R\mathcal{H}om(\mathcal{F},\mathcal{G}) \in D^b_c(X,\Bbbk)^{\ge 0}$

Pf: • Noetherian induction on X.
take stratification of X. w.r. $\mathcal{F}$ & $\mathbb{D}\mathcal{G}$ constr.

$j: U \underset{\text{open}}{\hookrightarrow} X \hookleftarrow Z : i$

Lem 3.1.6 $\Rightarrow$ $j^*\mathcal{F} \in D^b_{locf}(U,\Bbbk)^{\le -\dim U}$
$j^*\mathcal{G} \in D^b_{locf}(U,\Bbbk)^{\ge -\dim U}$

$\Rightarrow R\mathcal{H}om(j^*\mathcal{F}, j^*\mathcal{G}) \in D^b_c(U,\Bbbk)^{\ge 0}$

• distinguished triangle.

$i_*R\mathcal{H}om(i^*\mathcal{F}, i^!\mathcal{G}) \to R\mathcal{H}om(\mathcal{F},\mathcal{G}) \to j_*R\mathcal{H}om(j^*\mathcal{F}, j^*\mathcal{G}) \to$

j_* Right derived. $\Rightarrow$ RHS $\in D^b_c(X,\Bbbk)^{\ge 0}$

3.1.4 $\Rightarrow i^*\mathcal{F}$, $i^!\mathcal{G} \in {}^pD^b_c(Z,\Bbbk)^{\le 0}$

$\Rightarrow$ (induction) $R\mathcal{H}om(i^*\mathcal{F}, i^!\mathcal{G}) \in D^b_c(Z,\Bbbk)^{\ge 0}$

$\Rightarrow$ (3.1.4) $i_*R\mathcal{H}om(i^*\mathcal{F}, i^!\mathcal{G}) \in D^b_c(Z,\Bbbk)^{\ge 0}$

<u>Lem 3.2.2</u>. $\mathcal{L} \in \operatorname{Loc}^{ft}(X,\Bbbk) \Rightarrow$
$(-)\overset{L}{\otimes}\mathcal{L}$: right t-exact.
$R\mathcal{H}om(\mathcal{L},-)$: left t-exact w/. perverse t-structure
$\mathcal{L}$ locally free $\Rightarrow$ both t-exact.

(这是因为 $-\overset{L}{\otimes} j_s^*\mathcal{L}$ right exact.
for the natural t-str.
$\Rightarrow j_s^*\mathcal{F} \overset{L}{\otimes} j_s^*\mathcal{L} \in D^b_{locf}(X_s,\Bbbk)^{\le -\dim X_s}$.
& $\mathcal{L}$ locally free case $\Rightarrow -\overset{L}{\otimes} j^*\mathcal{L}$ t-exact.)

<u>Lemma 3.2.5</u>: $\mathcal{F} \in {}^pD^b_c(X,\Bbbk)^{\le 0}$. $\mathcal{G} \in {}^pD^b_c(Y,\Bbbk)^{\le 0}$
$\Rightarrow \mathcal{F} \boxtimes \mathcal{G} \in {}^pD^b_c(X \times Y, \Bbbk)^{\le 0}$

Moreover. $\Bbbk$ field $\Rightarrow \boxtimes$ t-exact. for the perv. t-str.

$(X_s)_{s\in\mathscr{S}}$ str. of X w/. $\mathcal{F}$
$(Y_t)_{t\in\mathscr{T}}$ str. of Y w/. $\mathcal{G}$
$\Rightarrow (X_s \times Y_t)$ str of $X \times Y$ w/. $\mathcal{F} \boxtimes \mathcal{G}$
$(\mathcal{F} \boxtimes \mathcal{G} \cong pr_1^*\mathcal{F} \overset{L}{\otimes} pr_2^*\mathcal{G})$

Lem 3.1.6 $\Rightarrow$
$\mathcal{F}|_{X_s} \in D^b_c(X_s,\Bbbk)^{\le -\dim X_s}$
$\mathcal{G}|_{Y_t} \in D^b_c(Y_t,\Bbbk)^{\le -\dim Y_t}$
$(\mathcal{F} \boxtimes \mathcal{G})|_{X_s \times Y_t} \cong (\mathcal{F}|_{X_s}) \boxtimes (\mathcal{G}|_{Y_t}) \in D^b_c(X_s \times Y_t, \Bbbk)^{\le -\dim(X_s \times Y_t)}$
$\Rightarrow \mathcal{F} \boxtimes \mathcal{G} \in {}^pD^b_c(X \times Y, \Bbbk)^{\le 0}$

$\mathcal{F}, \mathcal{G}$ perv. $\Bbbk$ field $\Rightarrow \mathbb{D}(\mathcal{F} \boxtimes \mathcal{G}) \cong (\mathbb{D}\mathcal{F}) \boxtimes (\mathbb{D}\mathcal{G}) \in {}^pD^b_c(X \times Y, \Bbbk)^{\le 0}$
$\mathcal{F} \boxtimes \mathcal{G} \in \operatorname{Perv}(X \times Y, \Bbbk)$

3.3 IC complex.

<u>Def</u>: $h: Y \hookrightarrow X$ locally closed embedding
Intermediate-extension functor
$h_{!*}: \operatorname{Perv}(Y,\Bbbk) \to \operatorname{Perv}(X,\Bbbk)$
by
$h_{!*}(\mathcal{F}) = \operatorname{im}({}^pH^0(h_!\mathcal{F}) \to {}^pH^0(h_*\mathcal{F}))$

(Ex 3.1.6)
$h: Y \hookrightarrow X$ locally closed embedding.
$\mathcal{F} \in \operatorname{Perv}(Y)$ $\mathcal{G} \in \operatorname{Perv}(X)$. $\mathcal{G}$ supp. on $\overline{Y} \setminus Y$
$\Rightarrow \operatorname{Hom}({}^pH^0(h_!\mathcal{F}), \mathcal{G}) = \operatorname{Ext}^1({}^pH^0(h_!\mathcal{F}), \mathcal{G}) = 0$
$\operatorname{Hom}(\mathcal{G}, {}^pH^0(h_*\mathcal{F})) = \operatorname{Ext}^1(\mathcal{G}, {}^pH^0(h_*\mathcal{F})) = 0$

$\operatorname{Hom}({}^pH^0(j_!\mathcal{F}), i_*P) = \operatorname{Hom}(i^*({}^pH^0(j_!\mathcal{F})), P)$
$= \operatorname{Hom}({}^pH^0(i^*j_!\mathcal{F}), P) = 0$

$\operatorname{Hom}(i_!P, {}^pH^0(j_*\mathcal{F})) = \operatorname{Hom}(P, i^!\,{}^pH^0(j_*\mathcal{F}))$
$= \operatorname{Hom}(P, {}^pH^0(i^!j_*\mathcal{F})) = 0$

<u>Lem 3.3.2</u>: $h: Y \hookrightarrow X$ Locally closed emb

(1) $\mathcal{F} \in \operatorname{Perv}(Y,\Bbbk)$. natural isom $h^*h_{!*}\mathcal{F} \cong \mathcal{F}$.

(2) $\mathcal{F} \in \operatorname{Perv}(Y,\Bbbk)$, $h_{!*}\mathcal{F}$ has no nonzero subobj / quot. supported on $\overline{Y} \setminus Y$

<u>Proof</u>: $h: Y \hookrightarrow \overline{Y}$ open embedding. h^* t-exact

$h^*h_{!*}\mathcal{F} = h^*\operatorname{im}({}^pH^0(h_!\mathcal{F}) \to {}^pH^0(h_*\mathcal{F}))$
$\cong \operatorname{im}(h^*\,{}^pH^0(h_!\mathcal{F}) \to h^*\,{}^pH^0(h_*\mathcal{F}))$
$\cong (h^*$ t-exact$)$ $\operatorname{im}({}^pH^0(h^*h_!\mathcal{F}) \to {}^pH^0(h^*h_*\mathcal{F}))$
$\cong \operatorname{im}(\mathcal{F} \xrightarrow{id} \mathcal{F}) = \mathcal{F}$

$Z = X \setminus Y$. $\quad i: Z \setminus X$ inclusion

$h_{!*}\mathcal{F}$ nonzero sub $\mathcal{G}$ supp. on Z

$\mathcal{G} \hookrightarrow h_{!*}\mathcal{F} \hookrightarrow {}^pH^0(h_*\mathcal{F})$.
by Ex. 3.1.6 $\mathcal{G} = 0$ □

<u>Lem 3.3.2.5</u>: $i: Z \hookrightarrow X \hookleftarrow Y : h$ w/. $X = \overline{Y}$.
$\mathcal{F} \in \operatorname{Perv}(X,\Bbbk)$ perverse sheaf
with no nonzero subobj. or quotients supp. on Z.
$\Rightarrow i^*\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\le -1}$.
$i^!\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\ge 1}$

Pf: $h_!h^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F} \to$. all in ${}^pD^b_c(X,\Bbbk)^{\le 0}$
$\Rightarrow \mathcal{F} = {}^pH^0(\mathcal{F}) \twoheadrightarrow {}^pH^0(i_*i^*\mathcal{F})$ surj.
no proper quot. $\Rightarrow {}^pH^0(i_*i^*\mathcal{F}) = 0$. i_* t-exact
$\Rightarrow i^*\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\le 1}$.
Samely $i^!\mathcal{F} \in {}^pD^b_c(Z,\Bbbk)^{\ge 1}$.

<u>Lem 3.3.3</u>: $h: Y \hookrightarrow X$
$h_{!*}: \operatorname{Perv}(Y,\Bbbk) \to \operatorname{Perv}(X,\Bbbk)$ fully faithful
$\mathcal{F} \in \operatorname{Perv}(Y,\Bbbk)$. $h_{!*}\mathcal{F}$ unique perverse sheaf on X s.t.
• supp. on $\overline{Y}$.
• $h_{!*}\mathcal{F}|_Y = \mathcal{F}$.
• no nonzero sub / quot. supp. on $\overline{Y} \setminus Y$

Pf: $\operatorname{Perv}^\circ(X,\Bbbk)$ full subcat. of $\operatorname{Perv}(X,\Bbbk)$.

$\mathcal{F}, \mathcal{G} \in \operatorname{Perv}^\circ(Y,\Bbbk)$ $\underset{0}{\operatorname{Hom}(i^*\mathcal{F}, i^!\mathcal{G})} \to \operatorname{Hom}(\mathcal{F},\mathcal{G}) \to \operatorname{Hom}(h^*\mathcal{F}, h^*\mathcal{G}) \to \underset{0}{\operatorname{Hom}(i^*\mathcal{F}, i^!\mathcal{G}[1])}$

$\Rightarrow \operatorname{Hom}(\mathcal{F},\mathcal{G}) \to \operatorname{Hom}(h^*\mathcal{F}, h^*\mathcal{G})$
$\Rightarrow h^*: \operatorname{Perv}^\circ(Y,\Bbbk) \to \operatorname{Perv}(Y,\Bbbk)$ fully faithful
essential surj: $\mathcal{F} \cong h^*h_{!*}\mathcal{F}$.
$\Rightarrow h^*$ equiv. of cat. □